

AWS Backup Cross-Region Replication for EC2 Summary Report

Objective

Configure AWS Backup to automatically back up an Amazon EC2 instance and replicate backups from US East (N. Virginia) to US East (Ohio) for disaster recovery.

Configuration Steps

1. Launched an EC2 instance in N. Virginia and added sample data.
2. Created PrimaryBackupVault in N. Virginia and DRBackupVault in Ohio.
3. Created an AWS Backup Plan with a daily backup rule and 35-day retention.
4. Enabled cross-region copy to DRBackupVault in Ohio.
5. Assigned the EC2 instance to the backup plan using the AWS Backup Default Service Role.
6. Executed an on-demand backup and verified successful backup and copy jobs.
7. Verified recovery points in both regions.

Benefits

Cross-region replication improves disaster recovery, protects against regional failures, enhances business continuity, increases data durability, and supports backup compliance requirements.

Issues Encountered

Initially, the backup failed due to an IAM permission error (ec2:DescribeTags). The issue was resolved by using the AWS Backup Default Service Role. Automatic copy verification required additional validation, after which the cross-region copy completed successfully.

Conclusion

The project successfully demonstrated automated EC2 backups and cross-region replication using AWS Backup. Recovery points were verified in both the primary and disaster recovery regions, ensuring a reliable backup and recovery solution.